

FitBuddy – AI Fitness Plan Generator

Project Planning Phase & Execution Roadmap Document

PROJECT TITLE
FitBuddy

CORE ENGINE
Gemini Models

DOCUMENT TYPE
Project Plan

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4.1 Phase Scope & Objectives

The **Project Planning Phase** defines the roadmap, timeline, resource allocation, risk evaluation, and sprint milestones necessary to build, test, and deploy **FitBuddy** using Google Gemini models.

Primary Objective

Establish an agile development workflow for seamlessly integrating Gemini API endpoints with an intuitive user interface.

Success Criteria

On-time milestone execution, zero prompt-schema breaking errors, and sub-3 second response time for generated fitness routines.

4.2 Work Breakdown Structure (WBS)

The development lifecycle is divided into structured, manageable work packages:

Module WBS	Task Breakdown	Deliverable Output
WBS 1.0: Environment	Set up repo, API credentials, SDK dependencies.	Configured Dev Environment
WBS 2.0: AI Pipeline	Design prompt templates & output schema validation.	Gemini Middleware Handler
WBS 3.0: UI Interface	Develop form inputs, loader, and schedule views.	Interactive Web Dashboard
WBS 4.0: QA & Deployment	Run API load tests, UI validation, and export checks.	Verified Release Build

4.3 Timeline & Milestone Schedule

Sprint 1: Setup & Prompt Tuning

Initialize project architecture and optimize Gemini model prompts for consistent JSON output generation.

Sprint 2: Application UI & State

Construct frontend forms for collecting user metadata (goals, equipment, duration) and rendering AI responses.

4.4 Resource & Technology Stack Allocation

Hardware, software, and external services required for the full implementation of FitBuddy:

Category	Technology / Resource	Purpose in FitBuddy
AI Model API	Google Gemini API (1.5 Flash / Pro)	Core AI engine for plan generation
Frontend	HTML5, CSS3, JavaScript / React	User interface & interactive plan viewer
Backend Middleware	Node.js / Python Fast API	API orchestration and error handling
Version Control	GitHub (Ravi-teja-777)	Codebase tracking and documentation

4.5 Risk Assessment & Mitigation Strategy

Identification of potential technical and operational risks during implementation along with countermeasures:

Risk Factor	Impact Level	Mitigation Strategy
API Rate Limits / Latency	High	Implement response caching, loading states, and fallback error handling.
Unstructured LLM Output	Medium	Enforce strict JSON response schemas and system instructions in prompt calls.
Invalid User Parameters	Medium	Implement frontend input validation before sending payloads to Gemini API.

4.6 Quality Standards & Deliverables Checklist

Core Deliverables - Functional Gemini Integration - User Input Validation Suite - Dynamic Weekly Plan Renderer - PDF Plan Export Tool	Completion Verification CompletedPhase plan approved CompletedTask backlog prioritized CompletedSDLC repository linked
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